

Alexandra McDaniel

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<https://github.com/alexhoppсалot/mcdaniel-portfolio.git>

PERSONAL PROFILE

Graduate student in Electrical Engineering (GPA 4.0) with research experience in power systems and underwater acoustics. Skilled in Python, MATLAB, and project management. Seeking opportunities to apply technical and analytical expertise to energy systems and reliability engineering.

EDUCATION

Master's in Electrical Engineering | Laramie, WY Expected Graduation: May 2026
University of Wyoming
GPA 4.0

Bachelor's in Applied Physics | Provo, UT July 2024
Brigham Young University
GPA 3.75

Rancho High School | Las Vegas, NV June 2018
Clark County School District
GPA 4.8 (weighted) | Valedictorian

WORK EXPERIENCE

Graduate Research Assistant | Laramie, WY August 2024 – Present
University of Wyoming Dept. of Electrical Engineering

- Analyzed power quality disturbances and their impact on renewable-integrated transmission systems, supporting ongoing research in grid reliability

Independent Research Contractor | Remote August 2023 – Present
Brigham Young University Hydroacoustic Lab

- Trained deep learning models for seabed classification using sound signatures from merchant ships

Knobles Scientific and Analysis
** Funded by the Office of Naval Research*

Field Technician | Laramie, WY May – August 2024
University of Wyoming Dept. of Botany

- Assisted with field surveys of vegetation and water potential in plants while leading the in-lab preparation and calibration of electrical field equipment

Undergraduate Research Assistant | Provo, UT January 2022 – July 2023
Brigham Young University Dept. of Physics and Astronomy

- Characterized a laboratory water tank for sound measurements with variability in the sound speed

Physics Lab Manager | Provo, UT January 2022 – December 2022
Brigham Young University Dept. of Physics and Astronomy

- Managed a team of six undergraduates and coordinated lab setup for eight courses, ensuring smooth delivery of experiments for 2,000+ students each semester

Physics Lab Technician | Provo, UT January – September 2019, May– December 2021
Brigham Young University Dept. of Physics and Astronomy

- Led the re-design of two introductory labs for undergraduate students

SKILLS

- Coding languages:
 - Proficient: Python, MATLAB
 - Intermediate: MATLAB Simulink, LabVIEW, Mathematica
 - Beginner: C++, CRBasic, R
- Software:
 - Proficient: RT-Lab, Github
 - Beginner: SolidWorks, Autodesk Inventor
- Management Experience:
 - Implementation of SCRUM and AGILE methodologies
 - Scheduling experience
 - Interviewing/hiring experience
 - Project management tools: Trello, ClickUp
- Bilingual: English / Spanish

PUBLICATIONS

- Hopps-McDaniel, Alexandra M., et al. "Deep sediment heterogeneity inferred using very low-frequency features from merchant ships." The Journal of the Acoustical Society of America 156.4 (2024): 2265-2274.
- Hopps-McDaniel, Alexandra M., and Tracianne B. Neilsen. "Temperature-induced sound speed variability in a laboratory water tank." Proceedings of Meetings on Acoustics. Vol. 51. No. 1. AIP Publishing, 2023.
- Harmer, Madeline, et al. "Leveraging scientific modeling to engage pre-med undergraduates in physics lab courses." Physical Review Physics Education Research 20.2 (2024): 020150.
- Undergraduate Capstone Project: Hopps, Alexandra, "Characterizing the sound speed variability in laboratory water tank," Advisor: Traci Neilsen (Capstone, Jun 2023).

PRESENTATIONS AT CONFERENCES

- May 2025 - Acoustical Society of America (New Orleans, Louisiana) | "The impact of water sound speed on deep learning seabed classification from shipping noise"
- May 2024 - Acoustical Society of America (Ottawa, Canada) | "Deep sediment characterization from very low-frequency features from merchant ships"
- May 2023 - Acoustical Society of America (Chicago, Illinois) | "Introducing sound speed variability in a laboratory water tank"
- July 2022 - American Association for Physics Teachers (Grand Rapids, Michigan) | "Physics labs that resonate with pre-medicine students"

RELEVANT COURSE WORK

- Engineering: Statistical Signal Processing, Power Reliability, Power Quality, Power Engineering, Communication Theory, Cyber Physical Systems, Thermodynamics, Biomedical Engineering
- Mathematics: Linear Algebra, Calculus 3, Ordinary Differential Equations, Partial Differential Equations, Theory of Statistics 1 & 2
- Programming: Convolutional Neural Networks, Python lab, MATLAB lab, C++
- Physics: Electricity and Magnetism 1 & 2, Acoustics, Mechanics, Optics, Modern Physics, Intro to Physics
- TA courses: Basic Circuits Lab, Communication Theory Lab, Introductory Physics Lab

VOLUNTEER EXPERIENCE

Victim Assistance Mentor | Provo, UT

June 2021 – December 2022

Utah Children's Justice Center

- Served children survivors of sexual violence and/or abuse by planning and then doing an activity with them for 2-3 hours each week